Food Process Technology 2

Science

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Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

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Food Process Technology 2 (A07826)

Short Title: Food Process Technology 2
Faculty Science and Computing
Department Science
Cluster Food Science
Author EDUGGAN
Credits 5 **Level:** Advanced

Description of Module / Aims

This module extends and deepens the students' knowledge of food processing technologies already taken at ordinary degree level. It emphasises the quantitative approach to various operating principles underlying food processing operations. Students obtain practical operating experience in a leading-edge biofunctional food processing environment.

Programmes

TECH-0039	BSc (Hons) in Food Science and Innovation (WD_SFSIN_B)	4 / 7 / M
TECH-0039	BSc (Hons) in Food Science and Innovation (WD_SFDSC_B)	4 / 1 / M

Indicative Content

- Materials balancing in standardisation of food components
- Heat transfer through materials
- Conduction, convection in heat exchangers; evaporation and drying
- Separation: centrifugal and filtration
- Viscosity in fluids; fluid flow; emulsification
- Plant design piping and instrumentation diagrams
- Evaporation & drying plant operation (evaporator, spraydryer, fluid bed and freeze-drying)
- Operation of heat exchangers (plate, scraped surface), homogenisation and microfluidisation
- Membrane separation processes (ultrafiltration, reverse osmosis and diafiltration)
- Use of unit processes in a case study outlining enrichment of a bioactive dairy product
- Processes relevant to different food industries (meat, cereals)

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Validate the utilisation of specialist equipment in a modern food processing facility.
2. Evaluate solutions to particular food processing strategies in product innovation.
3. Assess unit operations quantitatively in food processing.
4. Determine optimum parameters in the food processing industry.

Learning and Teaching Methods

- Lectures will cover the theory of the module, with tutorial-type classes to examine quantitative aspects.
- Lectures and practicals will be undertaken in conjunction with Teagasc Food Research Centres at Moorepark and Ashtown.

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	50%	3,4
Continuous Assessment	50%	
In-Class Assessment	50%	1,2

Assessment Criteria

<40%: Little or no demonstration of an understanding of the fundamentals of Food Process Technology.

40%-49%: Will be able to show limited understanding of the fundamentals of Food Process Technology.

50%-59%: As well as the above, the student will be to show reasonable understanding of the fundamentals of Food Process Technology.

60%-69%: In addition, the student will demonstrate more detailed knowledge of the all topics contained in Food Process Technology.

70%-100%: All of the above to excellent levels. In addition the learner will be able to demonstrate comprehensive knowledge and understanding of all learning outcomes.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	36	
Practical	12	
Independent Learning	87	

Essential Material

- Brennan, J.G. and A.S. Grandison. _Food Processing Handbook_. 2nd ed.. Weinheim: Wiley-VCH, 2012.
- Earle, R.L. _Unit Operations in Food Processing_. Web Edition. New Zealand: NZIFST, 2004.

Requested Resources

- Room Type: Computer Lab